Homework

Basics in linear algebra and numerical methods, 2025/26. semester I.

General instructions: Each week, you have to submit the solution of three arbitrary exercises. This should be done as a Python code. Comments should be written in English.

Scores: 3×3 per week; altogether: 54 in the whole semester.

Week 5

(a) Generate the matrix $A + S$, where $A = \begin{pmatrix} 5 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 5 \end{pmatrix}$ and S is a symmetric random matrix of size 3×3 (consisting uniformly distributed values in $[0, 1]$). Make 100 choices of S and plot the maximal eigenvalues of the matrix $A+S$. You may generate a random matrix S and change its components $S[2, 1], S[3, 1], S[3, 2]$ such that it becomes symmetric.

(b) Simulate a two-dimensional random walk such that a particle starts at the origin. In one time step, it jumps either right or left or upwards or downwards by one unit with probability 1/4. Run it over 1000 time steps and plot the distance of the particle from the origin.

(c) Use the framework in the last two examples in practice 9 to approximate the *sin* function with a third-order polynomial given by $a \cdot x^3 + b \cdot x^2 + c \cdot x + d$ on $[0, 2\pi]$ using an equidistant grid with 101 points. (The data set should be the sin data and the parameters the above a, b, c, d).

(d) For a continuous random variable $X \sim U(0, 1)$, we know that $E[f(X)] = \int_0^1 f(x)dx$ for any function f integrable. Use this nice property to approximate the value of the integral $\int_0^1 e^{x^2} dx$.
Hint 1: Recall if X is a random variable, then so is $f(X)$.
Hint 2: (Law of large numbers) recall we can approximate the expected value of any random variable Y by the sample average $\frac{1}{N} \sum_{j=1}^{N} Y_j$, where $Y_1, \ldots, Y_N$ are independent, identically distributed random variables.